INCH-POUND

MIL-C-48582B (AR)
21 December 90

SUPERSEDING
MIL-C-48582A (AR)
15 December 1987

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, TARGET POSITION BOARD, 11747198

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of target position printed circuit board assembly, hereinafter identified as "Target Position Board," a part of Range Computer Test Panel, Subassembly MX-8049/TPM-22, a part of Radar Test Set AN/TPM-22 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified. the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control, Materiel, General Specification Governing the Manufacture and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548169	Integrated Circuit
10548173	Integrated Circuit
10554938	Connector
11747198	Printed Wiring Board Assembly, Target Position Board
11747200	Schematic, Target Position Board

PACKAGING DATA SHEET

PS 11747198	Packaging of Target Position Board

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquistion functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 11747198, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the Target Position Board shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. Part identification and marking.

d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The Target Position Board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
5 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	O.O2 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ±10% per decade.
The 5-55-5 hertz sweep time is 7 minutes 35 seconds ±10%.

3.3.2 Shock. The Target Position Board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ±10%, 11 milliseconds ±10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

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3.3.3 Altitude.

3.3.3.1 Operating. The Target Position Board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Non-operating. The Target Position Board shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The Target Position Board shall meet the requirements of 3.4 and 3.5 when-stabilized thermally at any temperature between +32 degrees F and +131 degrees F.

3.3.4.2 Non-operating (storage) . The Target Position Board shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization) , and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating The Target Position Board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity. Each conductor shall be continuous electrically as specified on the schematic diagram 11747200.

3.4.2 Resistance. Resistance shall be as specified in Table I.

TABLE 1. Resistance

RESISTANCE	TERMINALS FROM	TO
1000 ohms ± 5%	P1-53	P1-30
1000 ohms ± 5%	P1-43	P1-30

3.5 Performance. When supplied with the inputs as specified in Table II, the Target Position Board shall yield the outputs as specified in Table II.

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TABLE II. Input and output characteristics

REQUIREMENT	CHARACTERISTIC CS	TERMINALS
Input Power	5.00 ± 0.25 Vdc regulated, AC ripple not greater than 50mvp-p, capacity 500ma.	P1-30,31,32(+) P1-24,25,26 (Ret.)
	-5.2 ± 0.1 Vdc regulated, AC ripple not greater than 50mvp-p, capacity 1.5A.	P1-9,10,11(-) P1-24,25,26 (Ret.)
Input Signals		
T_0	Positive pulse from 0 to 4.0 ± 1.0 Vdc, width of 0.79 ± 0.1 sec_t PRF of 20.0 ± 0.2 KHz.	As required (Sec. Fig. 4)
Signal #1 (Clock pulse)	120MHz ± 1KHz sine wave, 700 ± 100mvp-p into 50 ohm load.	J1
Signal #2 (Count-Up, Count Down)	50% square wave, from 0 to 4.0 ± 1.0 vdc, PRF of 250 ± 2 Hz.	P1-41, P1-39 & P1-24 (Ret.)
Target Clear	Positive pulse from 0 to 4.0 ± 1.0 Vdc, width of between 15.8 ± 0.2 us and 50.0 ± 0.5 us depending on Target Overflow output pulse, PRF of 20.0 ± 0.2 KHz on for 8.2 ± 0.1 sec. and off for 8.2 ± 0.1 sec. Positive going edge of Target Clear pulse is coincident with negative going edge of Target Overflow pulse.	P1-19 P1-25 (Ret.)
Strobe	Positive pulse from 0 to 4.0 ± 1.0 Vdc, width of 0.79 ± 0.10 us, PRF of 20.0 ± 0.2 KHz, delayed from system T_0 by 42.0 ± 5.2 US.	P1-33 P1-25 (Ret.)

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TABLE II. Input and output characteristics (Cont.)

REQUIREMENTS	CHARACTERISTICS	TERMINALS
Target Reset	Positive pulse from 0 to 4.0 ± 1.0 Vdc, width of 0.79 ± 0.10 us, PRF of 20 ± 0.2 KHz, leading edge coincides with leading edge of Target Clear Pulse.	P1-21 P1-25 (Ret.)
Output Signals		
2560M	50% square wave, from 0 V to 4.0 ± 1.0 Vdc period (T) of 16.384 ± 0.200 second(s)	P1-36 P1-24 (Ret.)
Carry	Negative pulse from 0 to 4.0 ± 1.0 Vdc, width of 2.00 ± 0.100 ms, T of 16.384 ± 0.200 s when signal #2 is applied to P1-39.	P1-47 P1-24 (Ret.)
Borrow	Negative pulse from 0 to 4.0 ± 1.0 vdc, width of 2.00 ± 0.100 ms, T of 16.38 ± 0.20 sec when signal #2 is applied to P1-41.	P1-49 P1-24 (Ret.)
Test Overflow Test #1	50% square wave, from 0 to 4.0 ± 1.0 Vdc, T of 34.160 ± 0.024 us when 0.0 ± 0.5 Vdc is applied to P1-19 & P1-21.	P1-17 P1-25 (Ret.)
Target Overflow Test #2	Positive pulse from 0 to 4.0 ± 1.0 Vdc, PRF of 20.0 ± 0.2 KHz, leading edge is delayed variably (pulse width oscillates between 10.0 ± 1.0 us and 27.0 ± 1.0 us with respect to Target Reset; each cycle takes 16.38 ± 0.2 sec between 23.0 ± 1.0 us and 40.0 ± 1.0 us from Target Reset when signal #2 is applied to P1-39, Target Reset to P1-21, and Target Clear to P1-19 (see Figure 1).	P1-17 P1-25 (Ret.)

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TABLE II. Input and output characteristics (Cont.)

REQUIREMENT	CH.ARACTERISTICS	TERMINALS
Target Overflow Test #3	Positive pulse from O to 4.0 ± 1.0 Vdc, PRF of 20.0 ± 0.2 KHz, leading edge is delayed variably (pulse width oscillates between 27.0 + 0.1 us and 10.0 ± 1.0 us with respect to Target Reset, each cycle takes 16.38 ± 0.2 see) between 40.0 + 1 us and 23.0 ± 0.10 us from Target Reset when signal #2 is applied to P1-41, Target Reset to P1-21, and Target Clear to P1-19 (see Figure 2a).	P1-17 P1-25 (Ret.)
Target Overflow Test #4	Positive pulse from O to 4.0 ± 1.0 Vdc, PRF of 20.0 ± 0.2 KHz, trailing edge is delayed 7.3 ± 0.3 us from System T_o, pulse width of 17.0 ± 1.0 us, when O + 0.5 Vdc is applied to P1-53, Target Reset to P1-21, and Target Clear to P1-19 (see Figure 2b).	P1-17 P1-25 (Ret.)
Target Overflow Test #5	Positive pulse from O to 4.0 ± 1.0 Vdc, PRF of 20.0 ± 0.2 KHz, trailing edge is delayed 34.1 ± 0.3 us from System T_o, pulse width of 17.0 ± 1.0 us, when O ± 0.5 Vdc is applied to P1-43, Target Reset to P1-21, and Target Clear to P1-19.	P1-17 P1-25 (Ret.)

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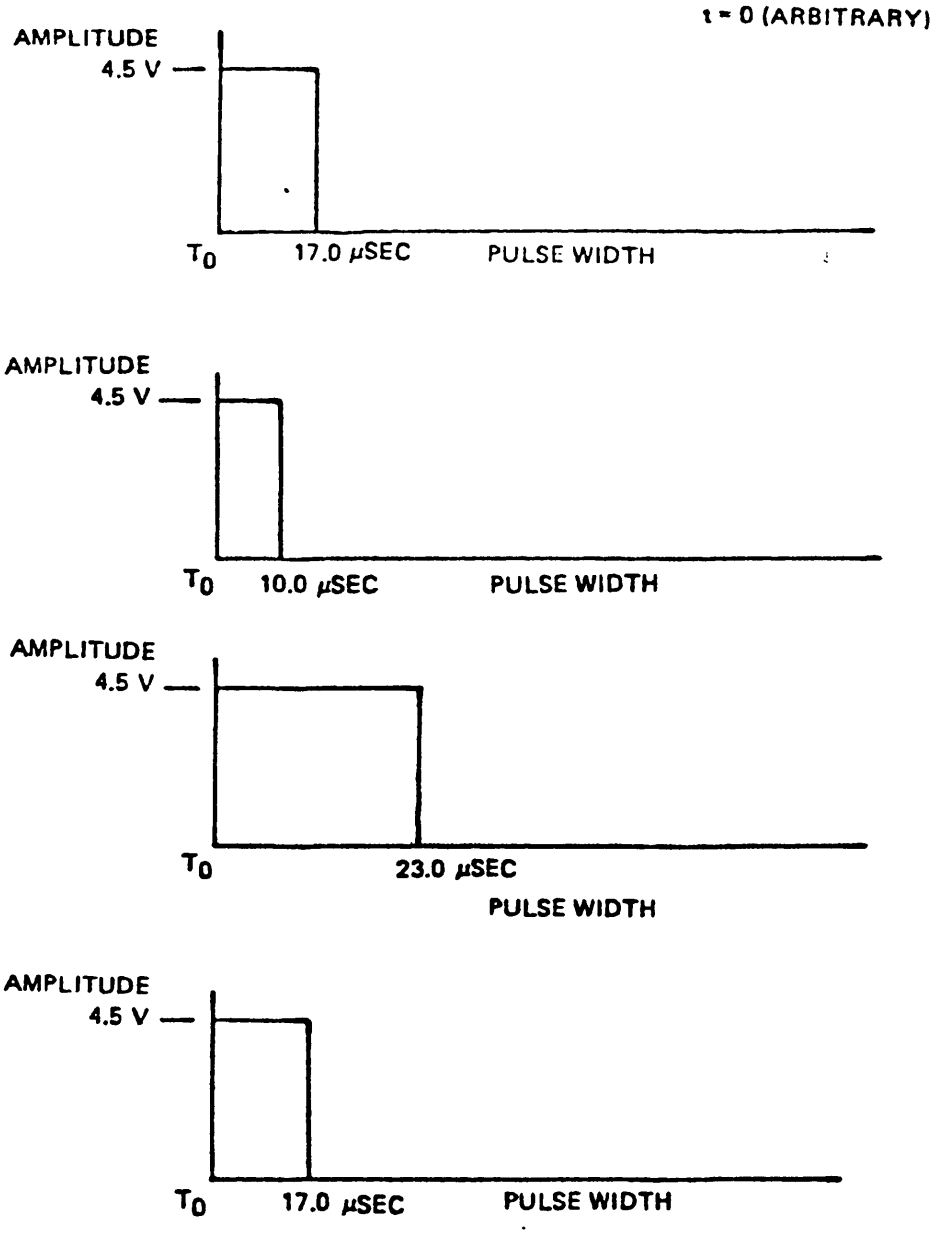


FIGURE 1. Target overflow test #2

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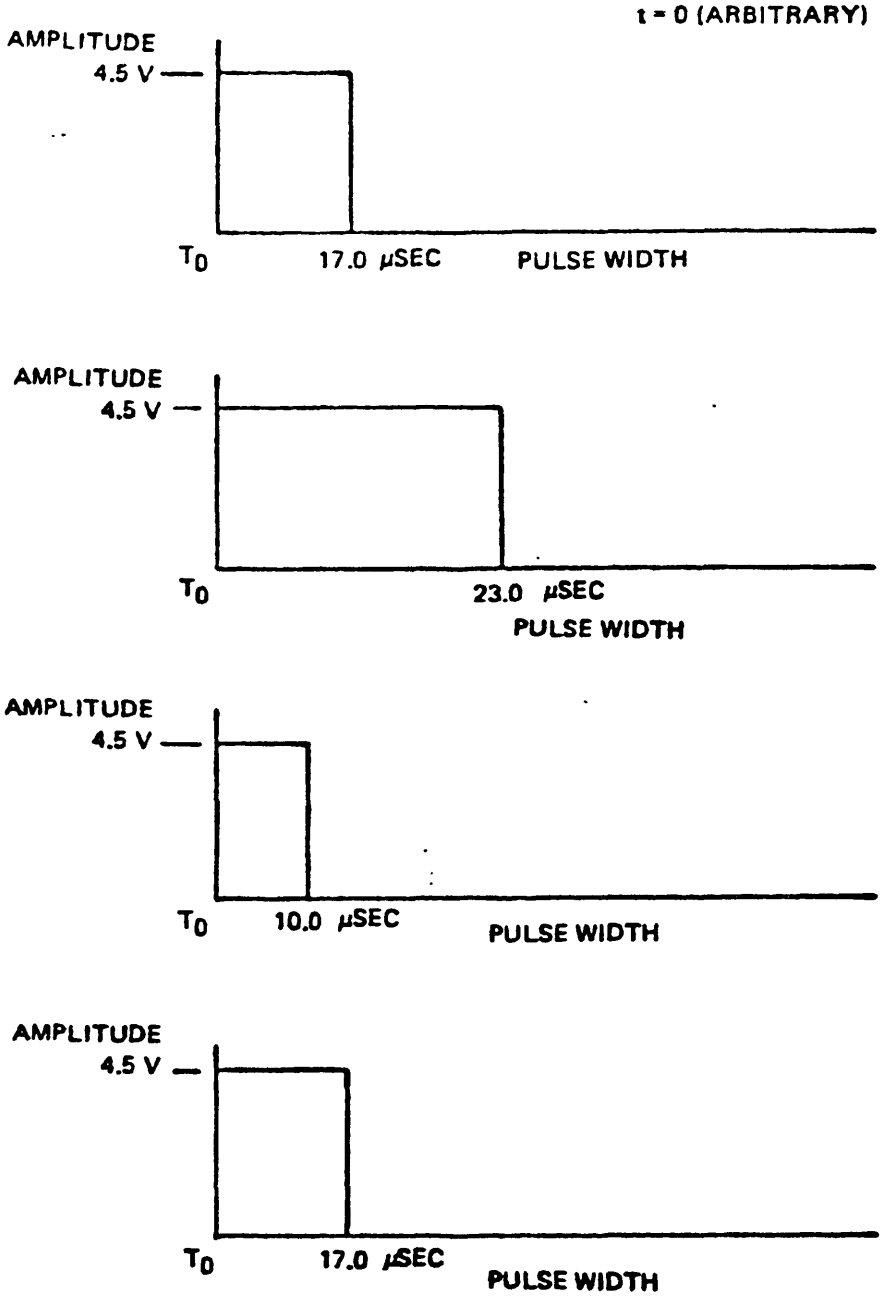


FIGURE 2a. Target overflow test #3

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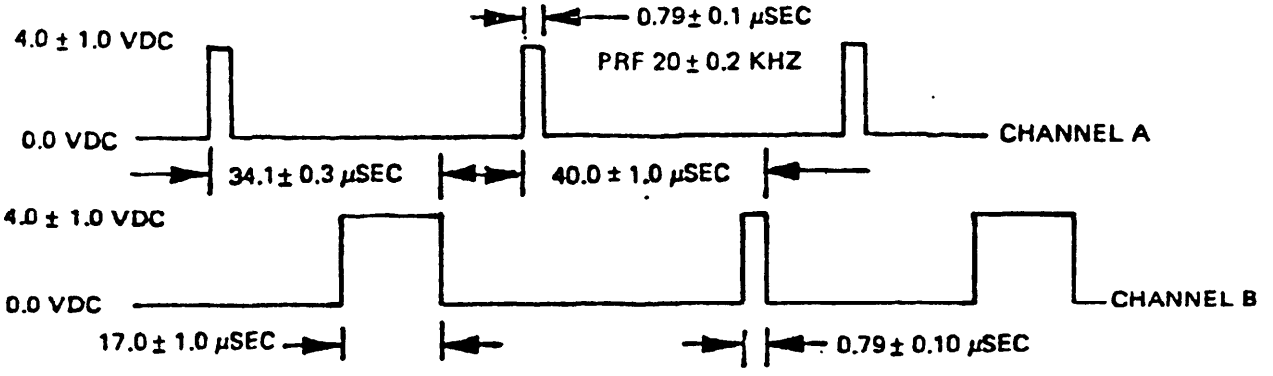


FIGURE 2b. Target overflow test # 4

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3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3) .

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of one Target Position Board for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed . First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table III and Table IV herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The government reserves the right to terminate its inspection upon any failure of any Target Position Board to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this-specification and the Quality Assurance Provisions (QAP) listed in the technical data package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with and the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in

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accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table III shall be performed on a 100% basis.

TABLE III. Functional tests
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Continuity	3.4.1	4.7.2
102. Resistance	3.4.2	4.7.2
103. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One Target Position Board out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table III and shall then meet the tests in Table IV. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE IV. Special sampling

CHARACTERISTIC	REQUIREMENTS	TEST PROCEDURE
Critical:	None	N/A
Major:		
104. Vibration	3.3.1	4.7.1.1
105. Shock	3.3.2	4.7.1.2
106. Altitude-operating	3.3.3.1	4.7.1.3
107. Altitude-nonoperating	3.3.3.2	4.7.1.3
108. Temperature	3.3.4	4.7.1.4
109. Humidity	3*3.5	4.7.1.5
Minor:	None	N/A

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4.5.1 Failure of any sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor–furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the “Methods of Inspection” specified in 4.7 of the specification and with the component inspection procedures specified in “Examination” and "Test Facilities” requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus , concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table V shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE V. Test equipment list

ITEM NUMBER	DESCRIPTION
1. Ohmmeter	Range 10 to 1000 ohms, accuracy ± 0.5%
2. Power supply	5.00 ± 0.25 Vdc regulated, ac ripple less than 50 mv p-p, capacity 500 MA.
3. Power supply	-5.2 ± 0.1 Vdc regulated, ac ripple less than 50 mV p-p, 1.5A capacity.
4. Sine-Wave Generator	120 MHz ± 1 KHz, 700 ± 100 mV p-p, into a 50 ohm load.
5. Oscilloscope	Dual channel; Rise time of scope plus probe not greater than 10 NSEC, Sweep period less than 50 NSEC. Input resistance: 8 megohms or greater.
6. Digital Voltmeter	0 to 10 Vdc, Accuracy 0.1%, input impedance 10 megohms or greater.
7. Test Fixture	As specified in Fig. 3. See Table VI for test fixture parts list.
8. Termination	500 ohms, ± 1%, 4 mw .

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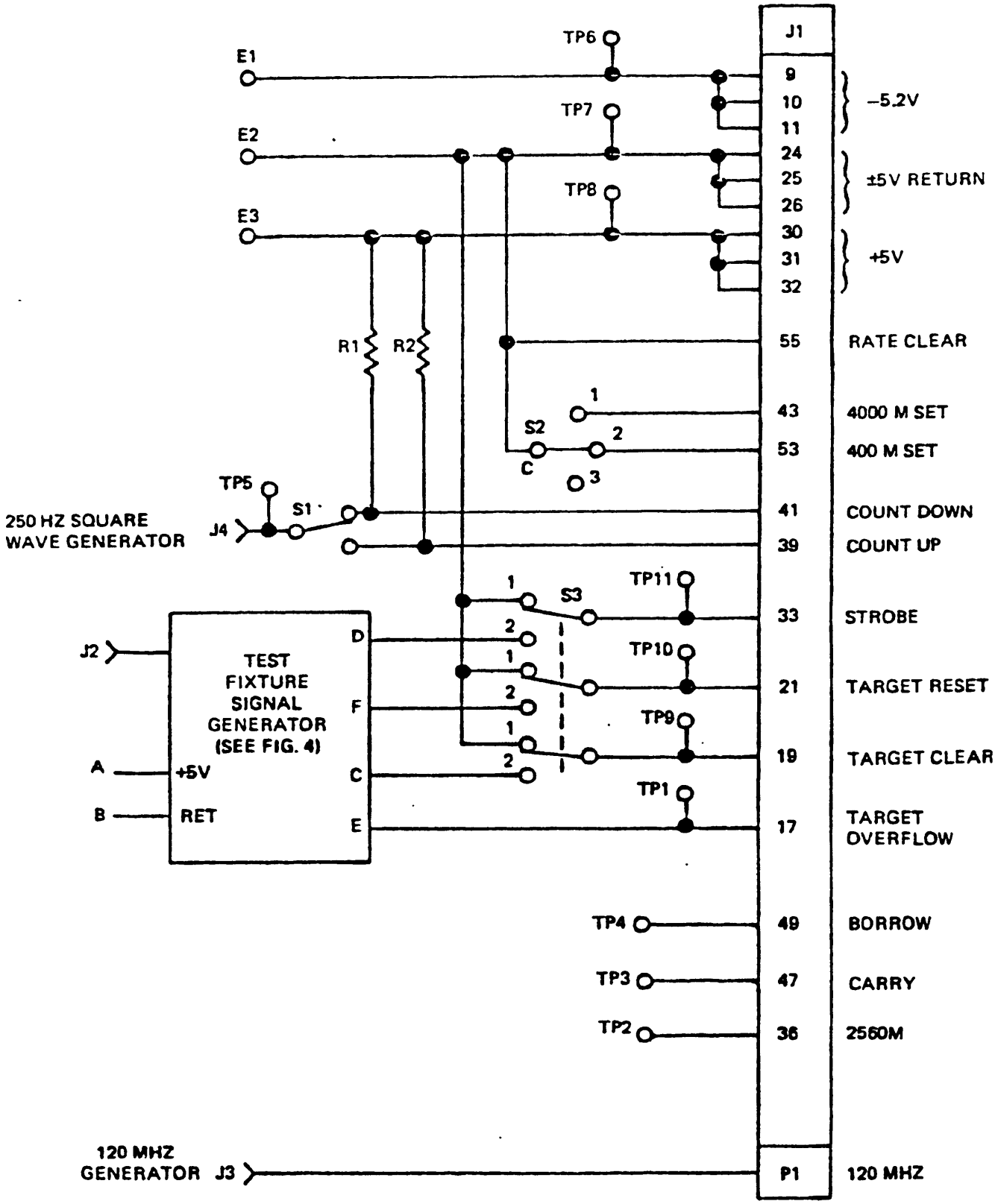


FIGURE 3. Test fixture setup

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TABLE VI. Test fixture parts list

DESIGNATION	NOMENCLATURE AND DESCRIPTION
A1	Test fixture signal generator in accordance with Figure 4. See Table VII for parts list.
J1	Connector, per 10554938 or equivalent
S1	Switch SPDT
S2	Switch SPTT
S3	Switch SPDT
TP1 thru TP11	Tip JACK, H.H. Smith 1501 or equivalent
J2, J3	Connector, BNC Receptacle
R1, R2	Resistor, 1K, 1/4 Watt 5%
P1	Connector, OSM 501-1
E1, E3	Binding post, H.H. Smith 257-R or equivalent
E2	Binding post, H.H. Smith 257-B or equivalent

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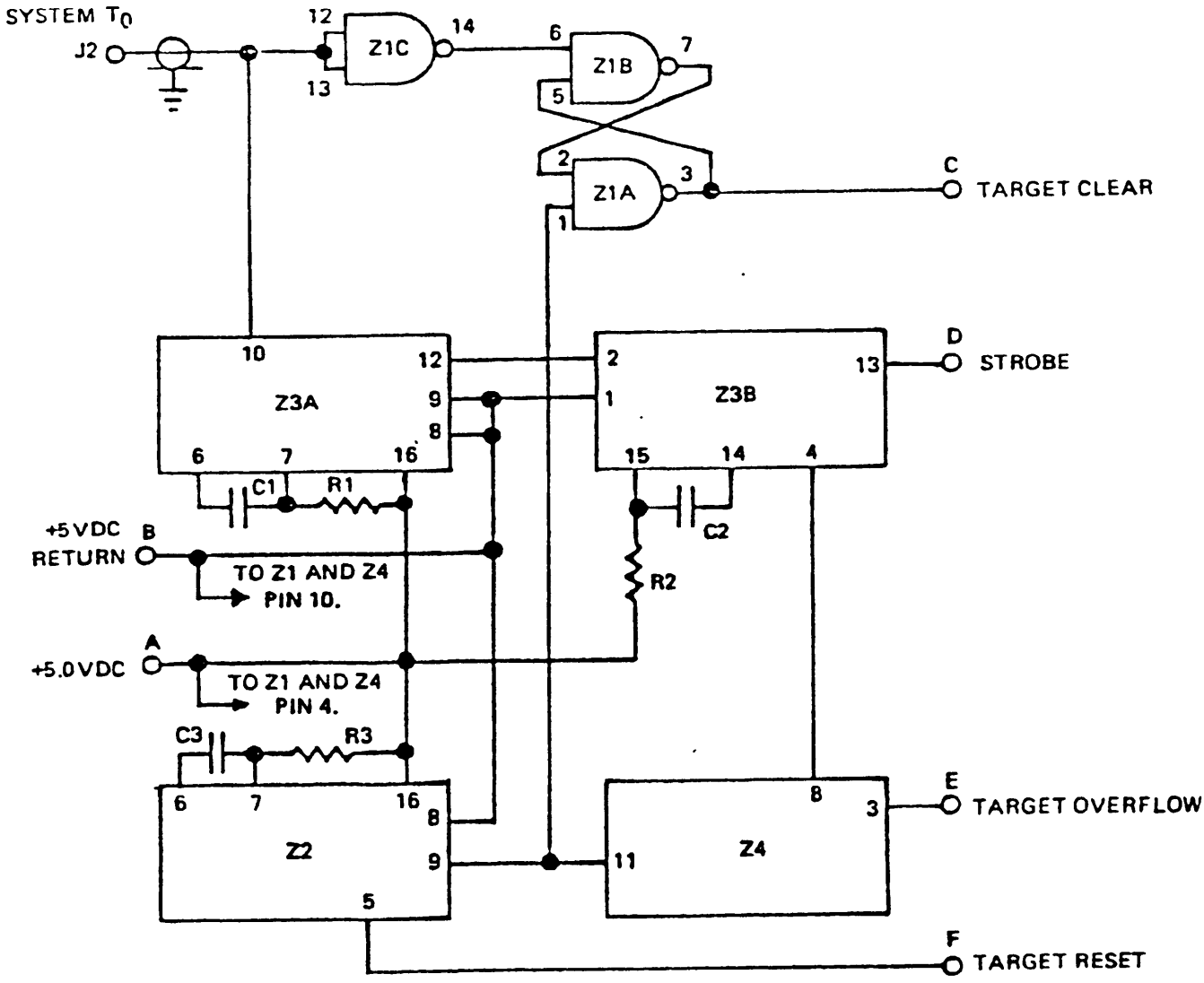


FIGURE 4. Test fixture signal generator

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TABLE VII. Test fixture signal generator parts list

DESIGNATION	NOMENCLATURE AND DESCRIPTION
C1	Capacitor 3900 pf ± 10%, 50 Vdc
C2, C3	Capacitor 330 pf ± 10%, 50 Vdc
R1	Resistor 33K ohms, ± 2%, 1/4 watt
R2, R3	Resistor 6.8K ohms ± 2%, 1/4 watt
Z1	I.C. RX3333 or SG140-37
Z2, Z3	I.C. 54123J
24	I.C. RX3371 or SF50-28

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of 11-2.1 apply. The Target Position Board shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the Target Position Board shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

co The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the Target position Board shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.3 for compliance with 3.3.3.

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4.7.1.4 Temperature.

4.701.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating) . This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1. Resistance shall be measured with an ohmmeter (Table V, item 1) for compliance with 3.4.2.

4.7.3 Performance. The selected Target Position Board shall be tested as specified in 4.7.3.1 through 4.7.3.1.3 to determine compliance with 3.5.

4.7.3.1 Test setup. Connect Target Position Board, Test Fixture (Table V, item 7) and Test Equipment (4.6.2.3) as shown in Figure 5. Unless otherwise specified, use external trigger on Oscilloscope (Table V, item 5).

4.7.3.1.1 Input power.

4.7.3.1.1.1 + 5 vdc. Use Digital Voltmeter (Table V, item 6) to measure and monitor + 5.00 ± 0.25 Vdc between TP8 & TP7 (Ret.) to determine compliance to + 5V Input Power requirement of Table II.

4.7.3.1.1.2 -5 Vdc. Use Digital Voltmeter (Table V, item 6) to measure and monitor -5.2 ± 0.1 Vdc between TP6 & TP7 (Ret.) to determine compliance to -5V Input Power requirement of Table II.

4.7.3.1.2 Signal #1 (clock pulse). Adjust Sine Wave Generator (Table V, item 4) to deliver 120 MHz + 1 KHz, 700 + 100 mv p-p into 50 ohms (Table V, item 8), sine wave To insure conformance to Signal #1 Input Signal requirement of Table II.

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4.7.3.1.3 Remaining input signals and output signals. The signals specified In Table VIII shall be observed sequentially (top to bottom of Table VIII) using the oscilloscope (Table V, item 5) at the specified test points and with the switch(s) set as specified to determine compliance with Input/Output signal requirements of Table II.

TABLE VIII. Performance test input/output signals

SIGNAL	TYPE	SWITCH# POSITION	TEST POINT
Signal #2	Input	------	TP5
Target Overflow Test #l (1)	output	S3-1	TP1
2560M	output	S1-1; S2-3	TP 2
Strobe	Input	S3-2	TP11
Borrow	output	S1-1; S2-3; S3-2	TP4
Target Clear	Input	S1-1; S2-3; S3-2	TP9
Target Reset	Input	S1-1; S2-3 ; S3-2	TP10
Target Overflow Test #3	output	S1-1; S2-3; S3-2	TP1
Carry	output	S1-2; S2-3; S3-2	TP 3
Target Overflow Test #2	output	S1-2; S2-3 ; S3-2	TP1
Target Overflow Test #5	output	S2-1; S3-2	TP1
Target Overflow Test #4	output	S2-2; S3-2	TP1

(1) Use normal trigger on Oscilloscope (Item 5.)

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

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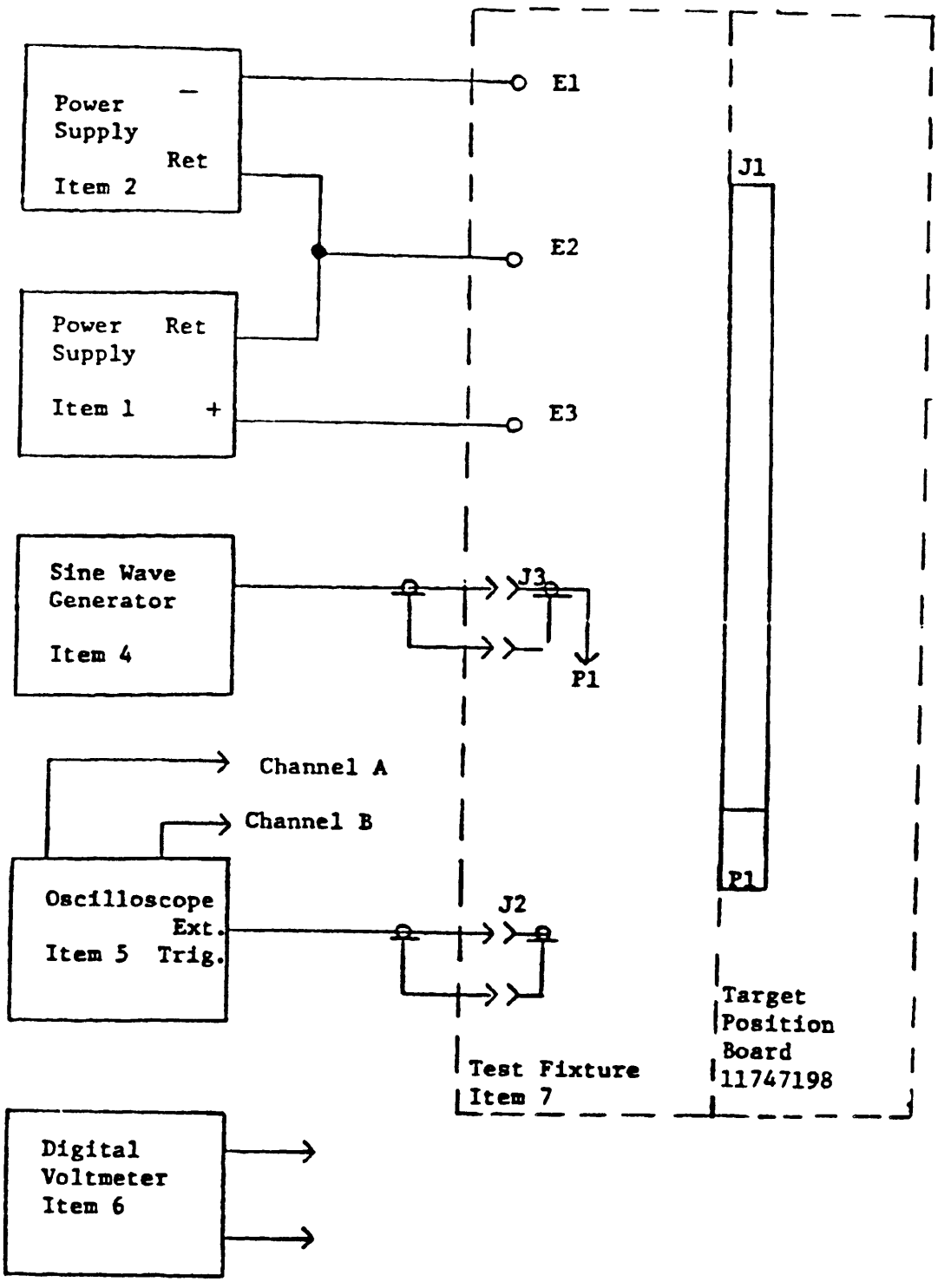


FIGURE 5. Target Position Test Setup

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5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS 11747198.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Target Position Board is used to provide a simulated target position signal to enable the Range Computer Tester, Subassembly MX-8094/TPM-22. to locate dynamically defective portions of unit 4 of Radar Set AN\VPS-2.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on-the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Radar Test Set AN/TPM-22
Test Panel Subassembly MX-8049/TPM-22

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6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A260)

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CIRCUIT CARD ASSEMBLY, TARGET POSITION BOARD, 11747198

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

[redacted]

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
US Army ARDEC Standardization & Specification Office	(201) 724-6662	880-6662

c. ADDRESS *(Include Zip Code)*

ATTN: SMCAR-BAC-S
Picatinny Arsenal, NJ 07806-5000

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